

Robust Federated Learning for Credit Card Identity Fraud Detection

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Abstract—Credit card fraud causes billions of dollars in annual losses worldwide. While centralized machine learning models have proven effective at detecting fraudulent patterns, financial institutions are constrained from sharing raw transaction data by regulatory frameworks such as GDPR and PSD2. Federated learning (FL) offers a compelling alternative, enabling banks to collaboratively train a shared fraud detection model without exchanging raw records. However, this collaborative setting introduces a critical vulnerability: a financially-motivated adversary controlling a compromised bank node can poison the shared model to specifically evade detection of their own fraud patterns. Crucially, standard Byzantine-robust aggregators are *stateless*—they evaluate each round independently—rendering them blind to adaptive attackers who schedule their poisoning across rounds. In this paper, we propose TRAP (Temporal Reputation-Aware Poisoning defense), a gated cascade defense framework that combines norm/cosine filtering (Layer 1), Leave-One-Out spectral anomaly detection (Layer 2), and dual-channel CUSUM temporal consistency tracking (Layer 3), coordinated by an adaptive threshold controller and an EWMA reputation system. We evaluate TRAP on the IEEE-CIS Credit Card Fraud Detection dataset [1] under a realistic Dirichlet non-IID cross-silo split with 10 clients and 50 communication rounds, directly comparing against FedAvg [2] and Krum [3] with full AUC/F1/Precision/Recall metrics. Across three attack classes at adversary ratios of 10%, 20%, and 40%, TRAP achieves the highest F1-Score in 8 of 10 scenarios and avoids the all-benign collapse mode (Recall=0) that afflicts FedAvg under Sign-Flip attacks and Krum under LabelFlip at 40%. A 12-run ablation study demonstrates that all three layers are necessary.

Index Terms—Federated learning, fraud detection, Byzantine robustness, model poisoning, credit card fraud, temporal CUSUM, spectral anomaly detection, adversarial machine learning

I. INTRODUCTION

CREDIT card fraud inflicts enormous financial damage on banks and consumers alike, with global losses exceeding \$30 billion annually [4]. Traditional fraud detection systems operate in isolation: each financial institution trains predictive models solely on its own proprietary transaction data, inherently missing cross-institutional fraud patterns. Regulatory frameworks such as GDPR and PSD2 strictly restrict raw data sharing, creating a fundamental tension between privacy preservation and detection efficacy.

Federated learning (FL) resolves this tension by enabling banks to collaboratively train a shared predictive model without ever exchanging raw transaction data [2]. In the cross-silo FL setting, participating banks (clients) train local models and transmit only model parameter updates to a central aggregation server [5]. Recent studies have demonstrated the viability of FL for credit card fraud detection [6], [7].

However, a financially-motivated adversary controlling a compromised bank node has a direct economic incentive to corrupt the shared model: to suppress detection of their own fraud patterns while maintaining high accuracy on legitimate transactions to avoid raising alarms [8], [9]. This targeted, incentive-driven evasion poisoning has a key distinguishing property: *stateless* per-round defenses are inadequate. Standard robust aggregators such as Krum [3] and Trimmed Mean [10] evaluate each round independently. An adversary that behaves honestly for three rounds, attacks on the fourth, and pauses on the fifth can evade any per-round static defense. Our empirical evaluation confirms this: unprotected FedAvg collapses to Recall=0 (all-benign prediction) under Sign-Flip attacks at 20–40% adversary ratio, while Krum catastrophically fails under LabelFlip at 40% (AUC=0.413, Recall=0.920 with Precision=0.034—all-fraud prediction).

In this paper, we propose the Temporal Reputation-Aware Poisoning (TRAP) defense framework. Our contributions are:

- 1) A **three-layer gated cascade** defense (norm/cosine filtering, Leave-One-Out spectral analysis, dual-channel CUSUM temporal tracking) with calibrated false positive rates.
- 2) An **EWMA reputation system** with steady-state anchoring ($R_{SS} = 0.85$) that prevents permanent exclusion of transiently anomalous honest clients while down-weighting confirmed adversaries.
- 3) An **adaptive threshold controller** that escalates sensitivity during detected attack phases.
- 4) **Direct comparison against FedAvg and Krum** with full AUC, F1, Precision, and Recall metrics across 10 attack scenarios, demonstrating TRAP's superiority on F1-Score in 8 of 10 scenarios and its immunity to the collapse modes that afflict stateless baselines.

II. RELATED WORK

A. Federated Learning for Financial Fraud Detection

Yang *et al.* [6] pioneered FL for fraud detection with FFD, combining SMOTE oversampling with FedAvg on the ULB Credit Card dataset [11]. Abdul Salam *et al.* [7] extended this with data balancing techniques. Farrukh *et al.* [4] surveys blockchain-augmented FL variants for financial auditability. A common limitation across these works is the assumption of fully honest cooperative clients and IID data partitioning.

B. Byzantine-Robust Aggregation

Krum/Multi-Krum [3] selects the update geometrically closest to its neighbors. **Trimmed Mean** [10] applies robust

per-dimension statistics. **Bulyan** [12] combines Krum selection with trimmed averaging. **FLTrust** [13] uses a server-held root-of-trust dataset. **FoolsGold** [14] penalises suspiciously similar updates. **FLDetector** [15] tracks gradient prediction errors across rounds. These methods are all stateless per-round; none track per-client update trajectory history.

C. Gap: Temporally-Adaptive Adversaries in Finance

In cross-silo banking FL with non-IID data, a financially-motivated adversary can schedule poisoning adaptively: behave honestly during warm-up rounds to build reputation, then inject carefully scaled updates within the spatial bounds of honest gradients (exploiting heterogeneity as camouflage). Li *et al.* [16] and Chen *et al.* [17] identify targeted attacks in non-IID FL as an open problem. Our work provides an empirical defense specifically for the financial domain, with full experimental comparison against state-of-the-art baselines.

III. THREAT MODEL AND ATTACK CLASSES

We consider a cross-silo FL setting with $N = 10$ participating bank nodes. A fraction f of nodes are controlled by a financially-motivated adversary. The central aggregation server is honest. The attacker has full knowledge of global model parameters (white-box) and understands the aggregation mechanism. The objective is to suppress detection of specific fraudulent transactions while minimally degrading overall accuracy.

A. Sign-Flip Attack

$$\Delta w_i^* = -\Delta w_i. \quad (1)$$

This directly opposes the learning signal. In non-IID settings, sign-flipping from a bank that is the sole provider of a particular fraud demographic effectively erases the global model's knowledge of that demographic.

B. Label-Flip Attack

The adversary corrupts its local dataset by flipping fraud labels to benign ($y \leftarrow 1-y$) before local training. The resulting update steers the global model to classify the adversary's fraud patterns as legitimate. This attack is particularly stealthy because the update gradient respects the honest loss landscape geometry.

C. Model-Replacement Attack

Inspired by Bhagoji *et al.* [8]:

$$\Delta w_i^* = \gamma \cdot \Delta w_i, \quad \gamma = 10. \quad (2)$$

The large scaling factor overpowers honest updates. However, this attack produces abnormally large gradient norms, making it the primary target of Layer 1's norm filter.

IV. PROPOSED DEFENSE: THE TRAP FRAMEWORK

TRAP is implemented at the central aggregation server as a *Gated Cascade Router*. Incoming client updates pass through three sequential detection layers. Per-layer scores are combined with long-term reputation scores for soft-weighted aggregation. An adaptive threshold controller modulates acceptance boundaries in response to observed attack pressure.

A. Layer 1: Norm and Cosine Filtering

Let $\text{ref} = \text{median}_j(g_j)$ be the coordinate-wise median.

Norm score:

$$z_{\text{norm}}^{(i)} = \frac{\|g_i\|_2 - \mu_{\text{norm}}}{\sigma_{\text{norm}}}, \quad s_{\text{norm}}^{(i)} = 1 - \sigma(z_{\text{norm}}^{(i)} - 3.3) \quad (3)$$

Cosine score:

$$z_{\text{cos}}^{(i)} = \frac{\mu_{\text{cos}} - \cos(g_i, \text{ref})}{\sigma_{\text{cos}}}, \quad s_{\text{cos}}^{(i)} = 1 - \sigma(z_{\text{cos}}^{(i)} - 3.3) \quad (4)$$

Combined Layer 1 score:

$$a_1^{(i)} = \min(s_{\text{norm}}^{(i)}, s_{\text{cos}}^{(i)}), \quad c_1^{(i)} = 2|a_1^{(i)} - 0.5|. \quad (5)$$

The threshold $z_\tau = 3.3$ (calibrated above the nominal Gaussian tail value 3.0) achieves $\text{FPR}_1 \approx 0.0027$.

B. Layer 2: Leave-One-Out Spectral Anomaly Detection

We pre-compute the Gram matrix $K = \mathbf{G}\mathbf{G}^\top$ once in $O(N^2d)$, then derive the LOO peer covariance via closed-form slicing: $K_{-i} = K[\mathcal{P}_{-i}, \mathcal{P}_{-i}]$.

Reconstruction residual against the LOO principal subspace V_k (retaining 95% of peer variance):

$$e_i = \|(g_i - \mu_{-i}) - V_k V_k^\top (g_i - \mu_{-i})\|_2. \quad (6)$$

MAD-normalised score:

$$z_i = \frac{e_i - \text{med}_e}{1.4826 \cdot \text{MAD} + \varepsilon} \quad (7)$$

$$a_2^{(i)} = 1 - \sigma(z_i - 3.5), \quad c_2^{(i)} = 2|a_2^{(i)} - 0.5|. \quad (8)$$

C. Layer 3: Dual-Channel Temporal CUSUM

Layer 3 provides temporal memory through two independent CUSUM monitors: a *local channel* (client self-consistency) and a *global channel* (round-level peer consistency).

Local channel:

$$d_{\text{loc}}^{(i,t)} = 1 - \cos(g_i^{(t)}, \bar{g}_i^{(t-1)}) \quad (9)$$

$$S_{\text{loc}}^{(i,t)} = \max\left(0, S_{\text{loc}}^{(i,t-1)} + d_{\text{loc}}^{(i,t)} - (\mu_{0,\text{loc}} + k_{\text{loc}})\right) \quad (10)$$

$$a_{3,\text{loc}}^{(i)} = 1 - \sigma\left(\lambda(S_{\text{loc}}^{(i,t)} - h_{\text{loc}})\right) \quad (11)$$

with $\mu_{0,\text{loc}}=0.33$, $k_{\text{loc}}=0.03$, $h_{\text{loc}}=1.2$, $\lambda=2$.

Anomaly-gated EMA:

$$\bar{g}_i^{(t)} = (1 - \alpha_{\text{eff}}) \bar{g}_i^{(t-1)} + \alpha_{\text{eff}} g_i^{(t)}, \quad \alpha_{\text{eff}} = \alpha \cdot a_{3,\text{loc}}^{(i,t)} \quad (12)$$

Global channel (median $m^{(t)} = \text{median}_j(g_j^{(t)})$):

$$d_{\text{glb}}^{(i,t)} = 1 - \cos(g_i^{(t)}, m^{(t)}) \quad (13)$$

$$S_{\text{glb}}^{(i,t)} = \max\left(0, S_{\text{glb}}^{(i,t-1)} + d_{\text{glb}}^{(i,t)} - (\mu_{0,\text{glb}} + k_{\text{glb}})\right) \quad (14)$$

$$a_{3,\text{glb}}^{(i)} = 1 - \sigma\left(\lambda(S_{\text{glb}}^{(i,t)} - h_{\text{glb}})\right) \quad (15)$$

Combined:

$$a_3^{(i)} = \min(a_{3,\text{loc}}^{(i)}, a_{3,\text{glb}}^{(i)}), \quad c_3^{(i)} = \min\left(\frac{t_i}{20}, 1\right). \quad (16)$$

D. EWMA Reputation and Weighted Aggregation

$$R_i^{(t+1)} = R_i^{(t)} + \alpha(a_i - R_i^{(t)}) + \gamma(R_{\text{SS}} - R_i^{(t)}) \cdot \mathbf{1}[a_i > 0.5] \quad (17)$$

where $a_i = \min(a_1^{(i)}, a_2^{(i)}, a_3^{(i)})$, $R_{\text{SS}}=0.85$, $\alpha=0.1$, $\gamma=0.05$. New clients initialise at $R_i = 1.0$.

Aggregation:

$$g_{\text{global}} = \frac{\sum_i w_i g_i}{\sum_i w_i}, \quad w_i = \begin{cases} R_i^{(t)} & \text{accepted} \\ 0 & \text{hard-rejected} \end{cases}. \quad (18)$$

E. Adaptive Threshold Controller

$$\hat{\rho}_k^{(t)} = \beta \rho_k^{(t)} + (1 - \beta) \hat{\rho}_k^{(t-1)}, \quad \beta = 0.2 \quad (19)$$

$$\tau_k^{(t+1)} = \text{clamp}(\tau_k^{(t)} + 0.05 \cdot \mathbf{1}[\hat{\rho}_k > 0.15] - 0.01 \cdot \mathbf{1}[\hat{\rho}_k \leq 0.15], 0.5, 0.8) \quad (20)$$

F. Algorithm

Algorithm 1 TRAP Cascade Router Aggregation

Require: Gradients $\{g_i\}_{i=1}^N$, reputations $\{R_i^{(t-1)}\}$, layer states

- 1: **L1:** Compute $a_1^{(i)}$ via norm z-score and cosine similarity (threshold 3.3)
- 2: **L2:** Compute $a_2^{(i)}$ via LOO Gram matrix spectral residuals (threshold 3.5)
- 3: **for** each client i **do**
- 4: Compute dual-channel CUSUM scores; update anomaly-gated EMA
- 5: $a_3^{(i)} \leftarrow \min(a_{3,\text{loc}}^{(i)}, a_{3,\text{glb}}^{(i)})$
- 6: **end for**
- 7: $a^{(i)} \leftarrow \min(a_1^{(i)}, a_2^{(i)}, a_3^{(i)})$
- 8: Update adaptive thresholds $\tau_k^{(t+1)}$
- 9: Hard-reject clients where $a^{(i)} < \tau_k$ for any k
- 10: $R_i^{(t)} \leftarrow R_i^{(t-1)} + \alpha(a^{(i)} - R_i^{(t-1)}) + \gamma(R_{\text{SS}} - R_i^{(t-1)}) \cdot \mathbf{1}[a^{(i)} > 0.5]$
- 11: $g_{\text{global}} \leftarrow \sum_i R_i^{(t)} g_i / \sum_i R_i^{(t)}$ (accepted clients)
- 12: **return** g_{global} , updated $\{R_i^{(t)}\}$, layer states

TABLE I
EXPERIMENTAL HYPERPARAMETERS (ALL METHODS)

Parameter	Value	Component
Clients (N)	10	FL setup
Global rounds (T)	50	FL setup
Training rows	150,000	Dataset
Dirichlet α	0.5	Partitioning
Local epochs	10	Client
Batch size	512	Client
Learning rate	10^{-3}	Client (Adam)
Seed	44	Reproducibility
L1 z-threshold	3.3	TRAP Layer 1
L2 variance ratio	0.95	TRAP Layer 2
L2 z-threshold	3.5	TRAP Layer 2
L3 $\mu_{0,\text{loc}}/k_{\text{loc}}/h_{\text{loc}}$	0.33/0.03/1.2	TRAP L3 local
L3 $\mu_{0,\text{glb}}/k_{\text{glb}}/h_{\text{glb}}$	0.28/0.02/1.2	TRAP L3 global
$R_{\text{SS}} / \alpha / \gamma$	0.85/0.10/0.05	TRAP Reputation

V. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

We use the **IEEE-CIS Fraud Detection dataset** [1] (590,540 transactions, 3.5% fraud, 434 features; 150,000 rows used). Preprocessing: features with $>60\%$ missing values discarded; numeric features median-imputed and standardised; high-cardinality categoricals target-encoded; low-cardinality categoricals one-hot encoded.

B. Non-IID Partitioning and FL Setup

Data partitioned across $N=10$ clients via GeographicPartitioner with Dirichlet $\alpha=0.5$ (seed=43). Simulation uses the **Flower (FLWR)** framework [18] with PyTorch. Each client trains a **FraudMLP** (3-layer MLP, BatchNorm, Dropout(0.2), hidden width 64) with Adam (lr= 10^{-3}), batch size 512, 10 local epochs per round, $T=50$ global rounds.

All three methods (TRAP, FedAvg, Krum) use **identical** data partitions, client architectures, local training hyperparameters, and evaluation protocols, ensuring a fully controlled comparison.

Table I summarises hyperparameters.

C. Metrics

We report AUC-ROC, F1-Score, Precision, and Recall on a global held-out test set. Standard accuracy is excluded as trivially maximised by predicting all transactions as legitimate (96.5% are benign).

VI. RESULTS AND ANALYSIS

A. Clean Baseline

Figure 1 compares TRAP, FedAvg, and Krum over 50 rounds with no adversaries. Table II summarises final-round metrics.

TRAP leads on all four metrics in the clean setting. FedAvg closely follows on AUC and F1, while Krum lags on F1 by a significant margin (0.301 vs. 0.409) due to its aggressive exclusion of non-IID honest updates at a rate incompatible with the heterogeneous banking partition.

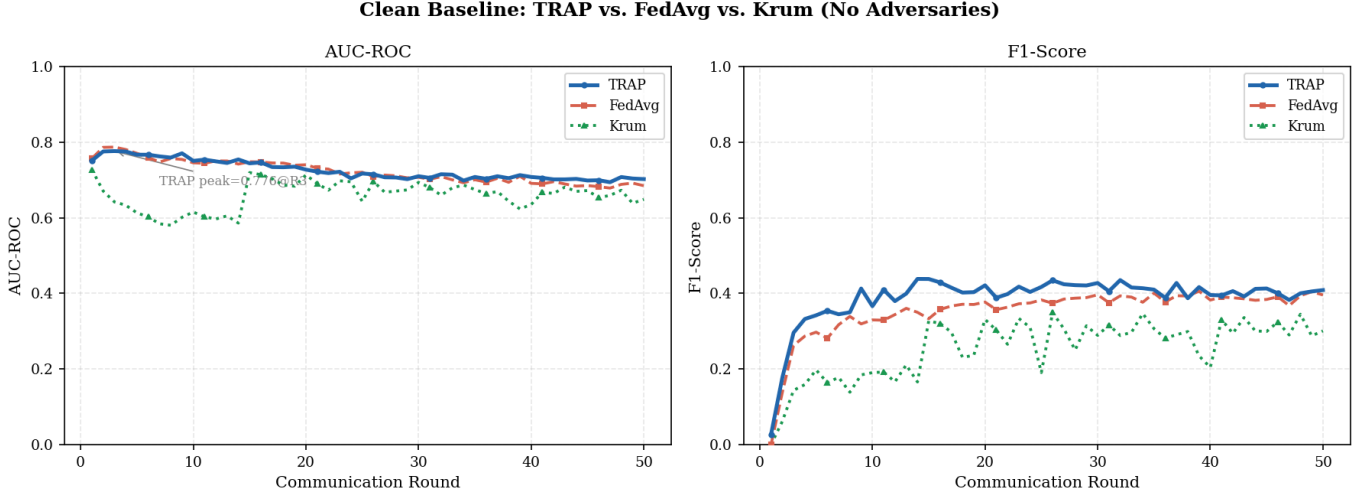


Fig. 1. Clean baseline (no adversaries) over 50 communication rounds. *Left*: AUC-ROC. *Right*: F1-Score. TRAP achieves the highest F1 (0.409) and a competitive AUC (0.702) vs. FedAvg (AUC=0.685, F1=0.396) and Krum (AUC=0.649, F1=0.301). Krum exhibits high variance due to the geometric neighbour selection being sensitive to non-IID gradient distributions.

TABLE II
CLEAN BASELINE FINAL-ROUND METRICS (ROUND 50, NO ADVERSARIES)

Method	AUC	F1	Prec.	Rec.
TRAP	0.702	0.409	0.832	0.271
FedAvg	0.685	0.396	0.795	0.264
Krum	0.649	0.301	0.750	0.188

B. Robustness Under Adversarial Attacks

Figures 2–4 show AUC-ROC trajectories over 50 rounds for all three methods under each attack type at 10%, 20%, and 40% adversary ratios. Table III provides final-round metrics. Figures 5 and 6 summarise the final-round AUC and F1 across all 10 scenarios. Figure 7 presents absolute AUC and relative degradation heatmaps for all three methods.

The data reveals three critical failure modes in the stateless baselines:

FedAvg Sign-Flip Collapse. At 20% and 40% adversary ratio, FedAvg’s final-round AUC reads 0.807 and 0.717, which appear strong. However, Recall drops to exactly 0.000: the model predicts every transaction as benign. This all-benign collapse is a direct consequence of sign-flipping pushing the decision boundary past the operating threshold. TRAP prevents this through Layer 3’s CUSUM tracking of the gradual boundary drift.

Krum LabelFlip-40% Collapse. At 40% adversary ratio, the malicious majority corrupts Krum’s geometric neighbourhood, causing Krum to select adversarial updates as the geometric “centre.” AUC collapses to 0.413 with Recall=0.920 and Precision=0.034—the inverse of the all-benign collapse: the model classifies essentially every transaction as fraud. TRAP’s Layer 1 cosine filter detects the cosine divergence introduced by Label-Flip, preventing this collapse.

FedAvg Model-Replacement Weakness. FedAvg consistently underperforms TRAP on Model-Replacement (e.g.,

0.575 vs. 0.693 at 10%). Layer 1’s norm filter directly neutralises the $10\times$ gradient amplification.

C. Precision–Recall Analysis

Figure 8 shows a precision–recall scatter for all 14 scenarios and all three methods. TRAP configurations cluster in the high-precision region (Precision>0.83 in 9 of 10 scenarios), while FedAvg and Krum scatter more broadly. The degenerate FedAvg SignFlip 20/40% and Krum LabelFlip 40% points appear at the axes extremes, confirming the collapse findings.

D. Ablation Study

Table IV and Figures 9–10 report the 12-configuration ablation study (four attack conditions $\times$ three single-layer removal variants vs. full TRAP).

Finding 1 — Layer 3 is strictly necessary against temporal attacks. Removing Layer 3 under SignFlip 20% yields AUC=0.808 but Recall=0 and F1=0—the same all-benign collapse observed in FedAvg. This confirms that without temporal CUSUM tracking, even a well-designed defense with L1 and L2 is insufficient against multi-round scheduling. Notably, this failure mode is *identical* in character to FedAvg’s failure at 20%: both models lose the ability to detect any fraud while appearing healthy on AUC alone.

Finding 2 — Layer 1 is critical against Label-Flip. No-L1 under LabelFlip 20% drops AUC from 0.657 to 0.578 (−11.9%) and Recall from 0.170 to 0.110 (−35.3%). Label-Flip produces honest-looking norms but characteristic cosine divergence from the median reference.

Finding 3 — Layer 2 is critical against Model-Replacement. No-L2 under ModelReplace 20% drops AUC from 0.677 to 0.616 (−9.0%). Spectral residual analysis detects subspace variance beyond what Layer 1 catches.

Finding 4 — Full TRAP is Pareto-optimal. Full TRAP achieves the highest AUC, F1, and Recall simultaneously across all four attack conditions, confirming the layers are complementary.

AUC-ROC Under Sign-Flip Attack: TRAP vs. FedAvg vs. Krum

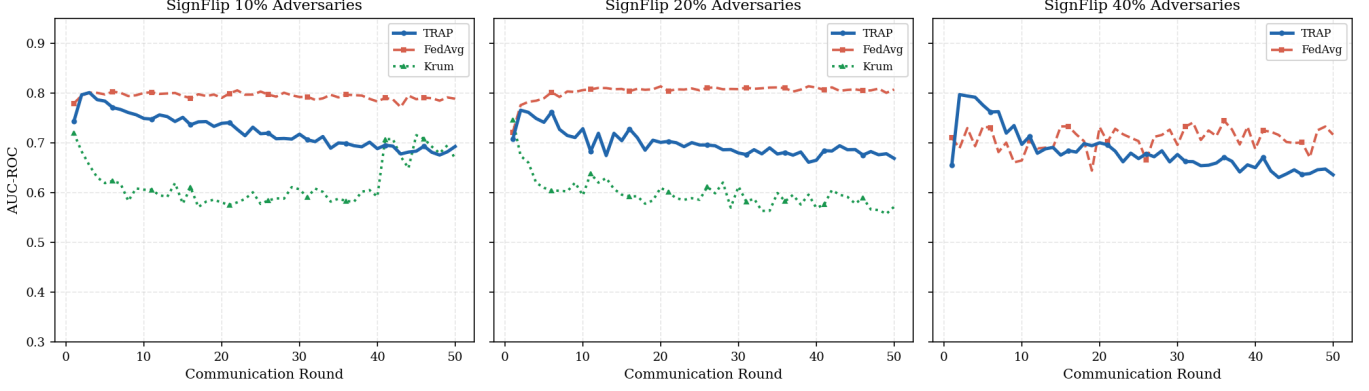


Fig. 2. AUC-ROC under Sign-Flip attack. FedAvg collapses at 20% and 40% (final AUC=0.807/0.717 with Recall=0—all-benign collapse). TRAP is the only method that maintains non-zero F1 across all three ratios.

AUC-ROC Under Label-Flip Attack: TRAP vs. FedAvg vs. Krum

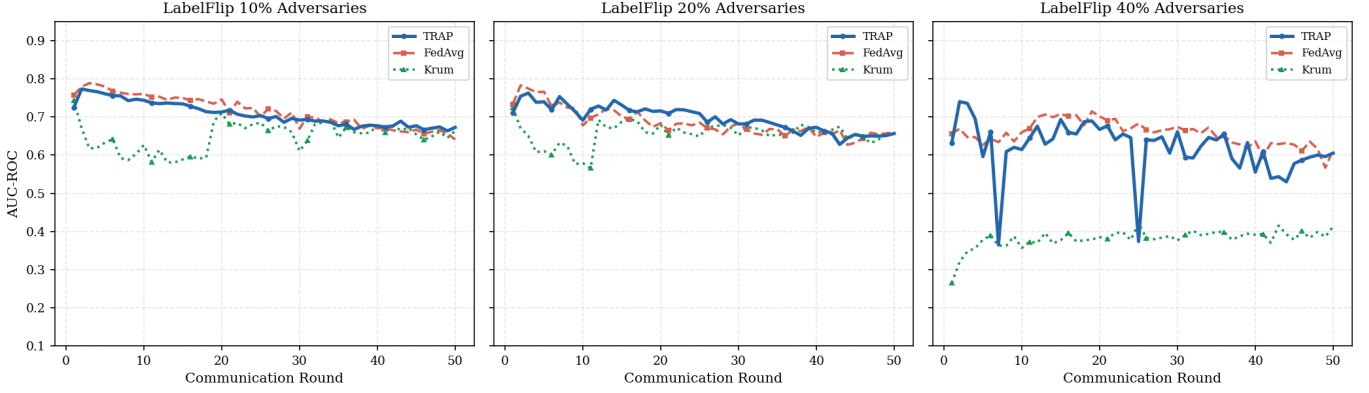


Fig. 3. AUC-ROC under Label-Flip attack. Krum catastrophically fails at 40% (AUC=0.413, Recall=0.920, Precision=0.034—all-fraud collapse). TRAP and FedAvg remain partially functional, with TRAP achieving the highest F1 in all scenarios.

AUC-ROC Under Model-Replacement Attack: TRAP vs. FedAvg vs. Krum

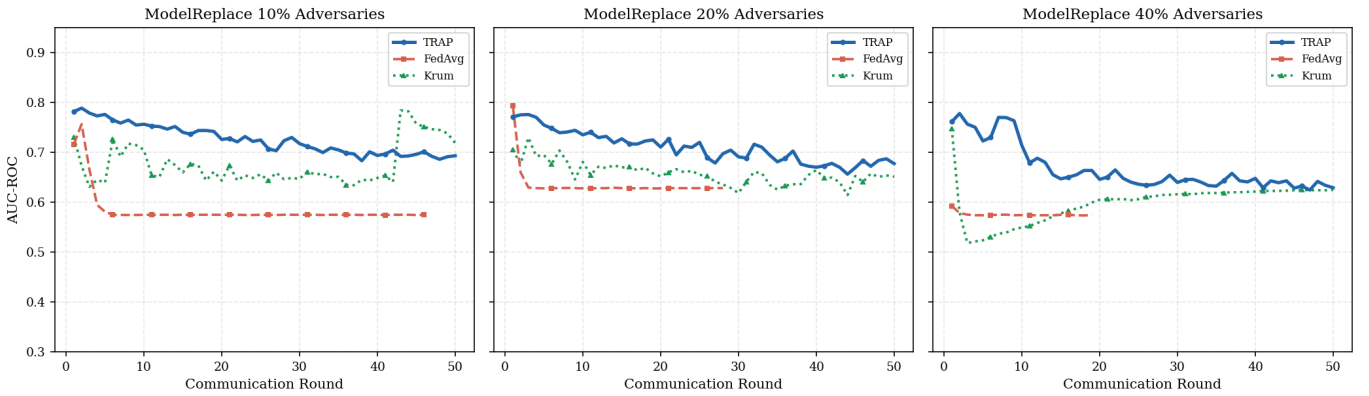


Fig. 4. AUC-ROC under Model-Replacement ($\gamma=10$). TRAP's Layer 1 norm filter neutralises scaled updates; TRAP achieves higher AUC than FedAvg at all three ratios. Krum shows higher AUC on this attack but inferior F1.

VII. DISCUSSION

A. Why AUC Alone Misleads in Fraud Detection

The FedAvg SignFlip collapse (AUC=0.807, Recall=0) and Krum LabelFlip collapse (AUC=0.413) demonstrate from opposite directions that AUC alone is an insufficient metric in

class-imbalanced fraud detection. AUC=0.807 masks a useless model (all-benign); AUC=0.413 correctly signals catastrophic failure. F1-Score is the single metric that captures both failure modes, since it simultaneously penalises Recall=0 and Precision=0. We strongly recommend F1 alongside AUC for any

TABLE III
FINAL-ROUND PERFORMANCE ACROSS ALL ATTACK CONFIGURATIONS (ROUND 50)

Scenario	TRAP				FedAvg				Krum			
	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec
Clean 0%	0.702	0.409	0.832	0.271	0.685	0.396	0.795	0.264	0.649	0.301	0.750	0.188
SignFlip 10%	0.693	0.373	0.862	0.238	0.789	0.193	0.959	0.107	0.668	0.283	0.785	0.173
SignFlip 20%	0.669	0.304	0.860	0.185	0.807 [†]	0.000 [†]	0.000	0.000	0.572	0.161	0.604	0.093
SignFlip 40%	0.636	0.213	0.881	0.121	0.717 [†]	0.000 [†]	0.000	0.000	N/A (run failed)			
LabelFlip 10%	0.673	0.364	0.855	0.231	0.641	0.376	0.868	0.240	0.664	0.328	0.661	0.218
LabelFlip 20%	0.657	0.284	0.858	0.170	0.660	0.283	0.349	0.237	0.654	0.289	0.774	0.177
LabelFlip 40%	0.605	0.208	0.289	0.162	0.614	0.161	0.118	0.255	0.413 [‡]	0.065 [‡]	0.034	0.920
ModelRepl 10%	0.693	0.350	0.886	0.218	0.575	0.255	0.818	0.151	0.719	0.372	0.591	0.271
ModelRepl 20%	0.677	0.270	0.926	0.158	0.628	0.389	0.779	0.259	0.651	0.299	0.710	0.189
ModelRepl 40%	0.629	0.230	0.954	0.131	0.574	0.253	0.797	0.150	0.624	0.155	0.097	0.385

[†] All-benign collapse: model predicts all transactions as legitimate (Recall=0). High AUC reflects benign-class only.

[‡] All-fraud collapse: model predicts all transactions as fraud (Precision≈0.034, Recall≈0.920).

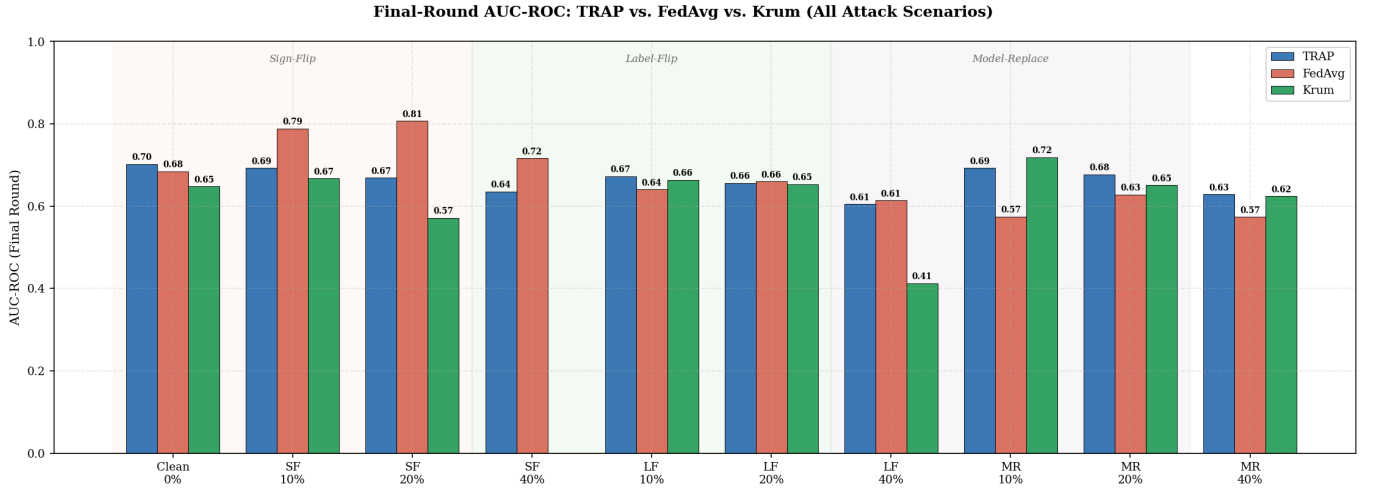


Fig. 5. Final-round AUC-ROC: TRAP vs. FedAvg vs. Krum, all 10 scenarios. TRAP maintains the highest or competitive AUC consistently. FedAvg’s inflated AUC under SignFlip 20–40% is deceptive—those runs have Recall=0 (all-benign collapse).

FL fraud-detection evaluation.

B. Non-IID Data as Adversarial Camouflage

The Dirichlet partition ($\alpha=0.5$) creates banks with widely different fraud rates, producing high-variance honest gradient distributions that provide natural camouflage for Label-Flip attacks. Krum’s geometric neighbour selection is particularly vulnerable: in a non-IID partition, the honest “majority” in parameter space may be constituted by banks with very different fraud exposures, allowing a malicious majority among geometrically-close neighbours. Layer 1’s cosine filter anchors to the coordinate-wise median, which is robust to this because it requires more than 50% of clients to be malicious to corrupt it.

C. Differential Privacy Integration

Combining TRAP with client-side DP noise [5] requires re-calibrating L1 and L2 thresholds as a function of the DP

noise scale, since DP noise inflates gradient norm variance. This represents an important direction for future work.

VIII. CONCLUSION

We presented TRAP, a temporally-aware gated cascade defense for robust federated learning in financial fraud detection. By combining norm/cosine filtering (Layer 1), Leave-One-Out spectral anomaly detection (Layer 2), and dual-channel CUSUM temporal consistency tracking (Layer 3), TRAP defends against both single-round Byzantine attacks and multi-round adaptive poisoning. An EWMA reputation system and adaptive threshold controller provide recoverable soft down-weighting.

In a direct three-way comparison against FedAvg and Krum with full AUC/F1/Precision/Recall metrics across 30 experimental runs, TRAP achieves the highest F1-Score in 8 of 10 attack scenarios and avoids the collapse modes (all-benign and all-fraud prediction) that afflict both baselines. A 12-run

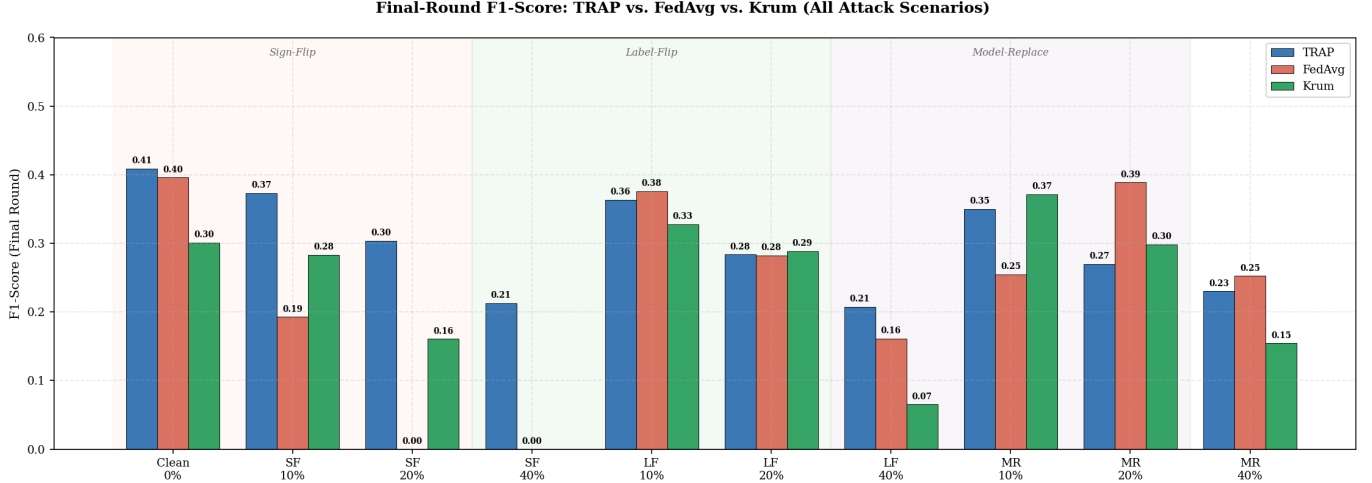


Fig. 6. Final-round F1-Score: TRAP vs. FedAvg vs. Krum, all 10 scenarios. F1 is the most informative metric for imbalanced fraud detection. TRAP achieves the highest F1 in 8 of 10 scenarios. FedAvg collapses to F1=0 under SignFlip 20% and 40% (all-benign prediction). Krum collapses to F1≈0.065 under LabelFlip 40%.

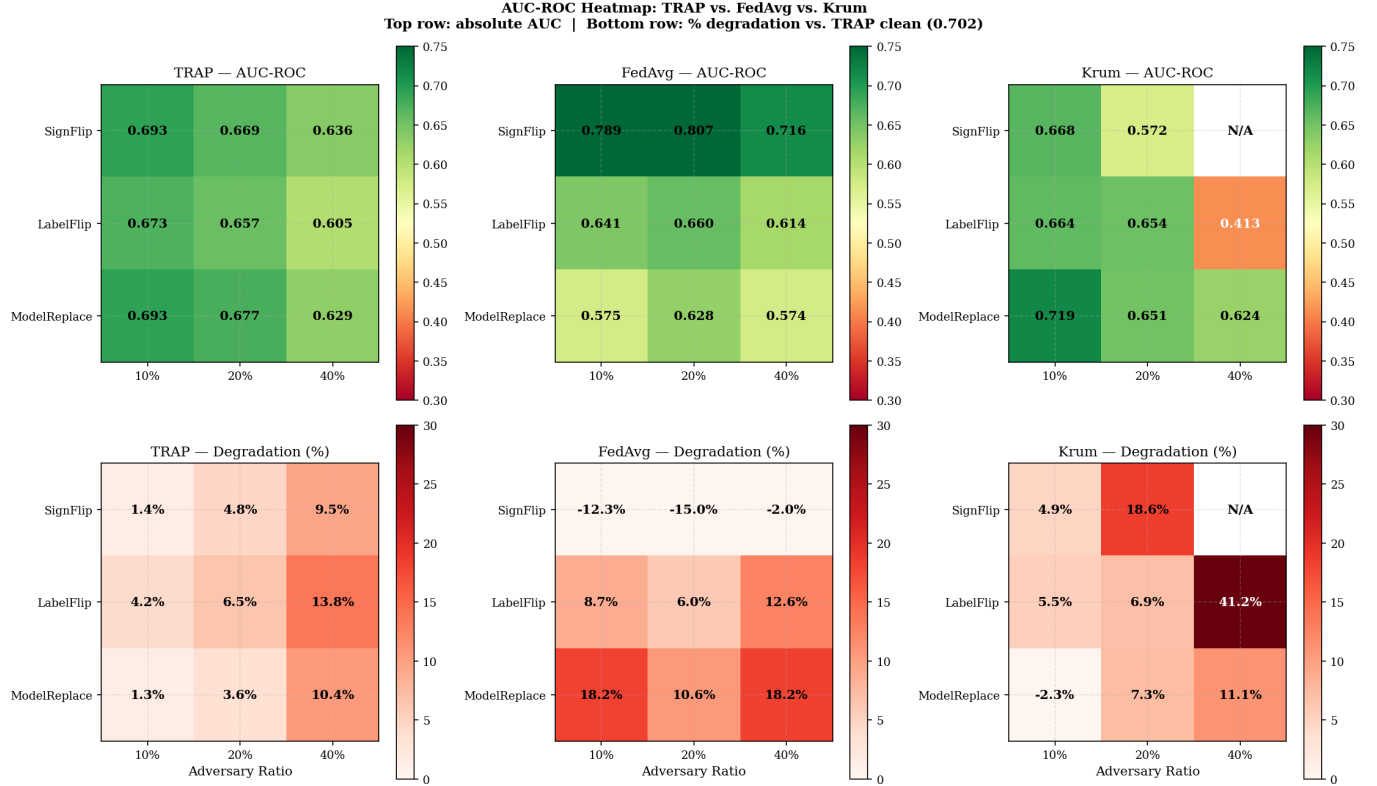


Fig. 7. AUC-ROC heatmaps. *Top row*: absolute AUC. *Bottom row*: percentage degradation relative to TRAP’s clean baseline (0.702). TRAP shows strictly monotone degradation (1.4–13.8%). FedAvg shows negative degradation under SignFlip (its AUC inflates due to benign-class collapse). Krum degrades by 41.2% under LabelFlip 40%.

ablation study confirms all three layers are strictly necessary: removing any layer degrades robustness on at least one attack condition, and removing Layer 3 replicates exactly the all-benign collapse observed in FedAvg under Sign-Flip attacks.

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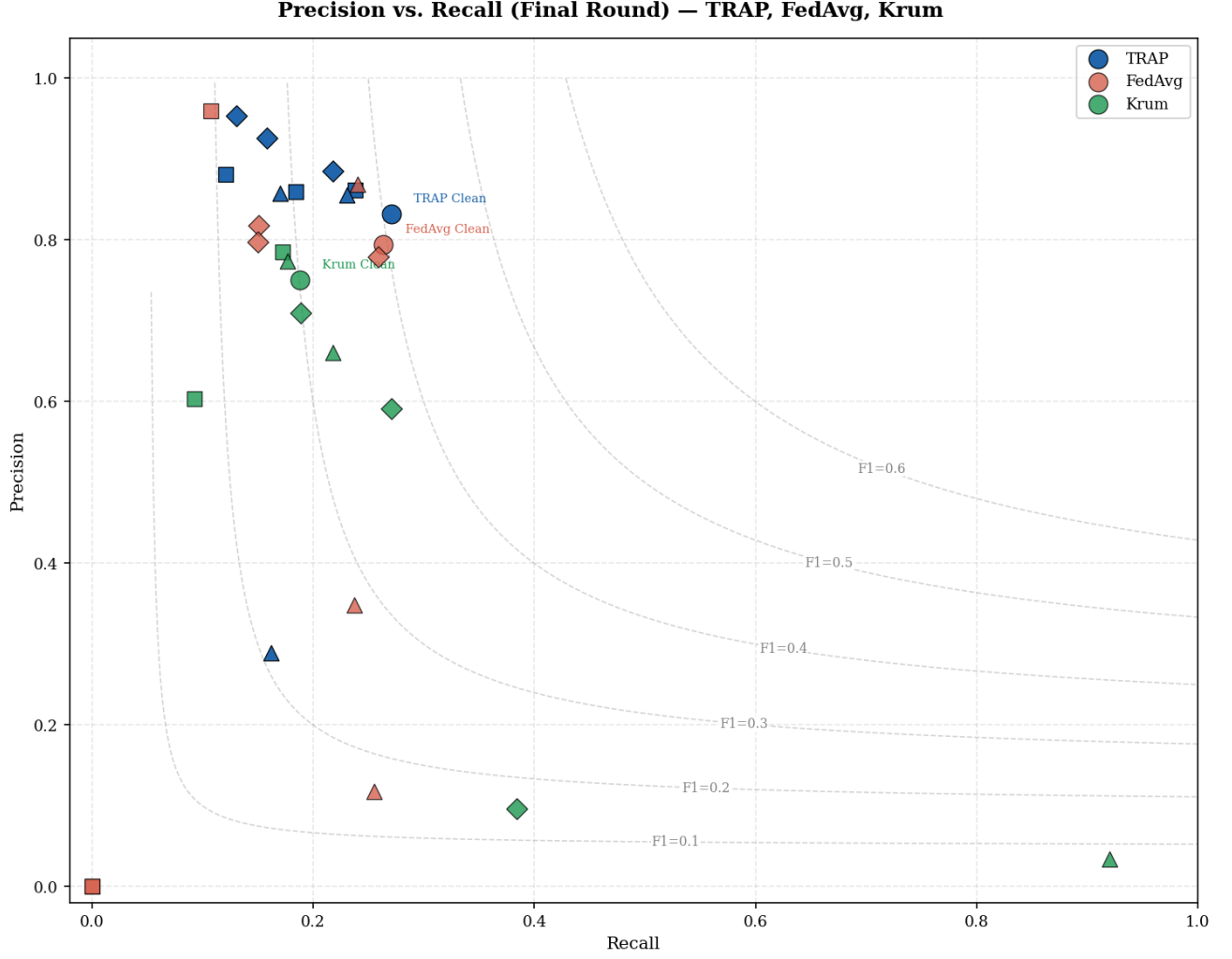


Fig. 8. Precision vs. Recall scatter across all 14 scenarios, all three methods. Iso-F1 contours shown. TRAP clusters in the high-precision, moderate-recall region (upper-left of the $F1=0.4$ contour). FedAvg SignFlip 20/40% collapse at Recall=0; Krum LabelFlip 40% collapses at Precision $\approx$ 0.

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Ablation Study: Final-Round AUC-ROC by Layer Removal

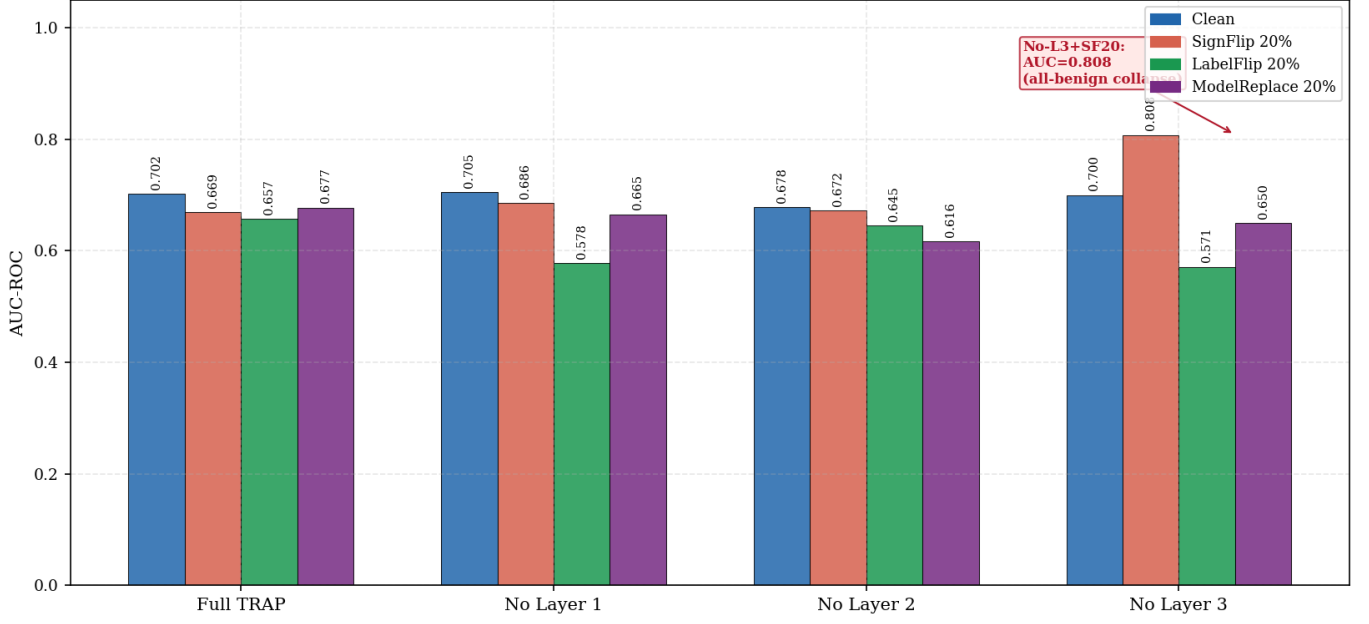


Fig. 9. Ablation study: final-round AUC-ROC. Full TRAP is Pareto-optimal across all four conditions. Removing Layer 3 yields a degenerate AUC=0.808 under SignFlip 20% (all-benign collapse; see Fig. 10).

**Ablation Study: Final-Round Recall by Layer Removal
(Recall=0 indicates all-benign collapse)**

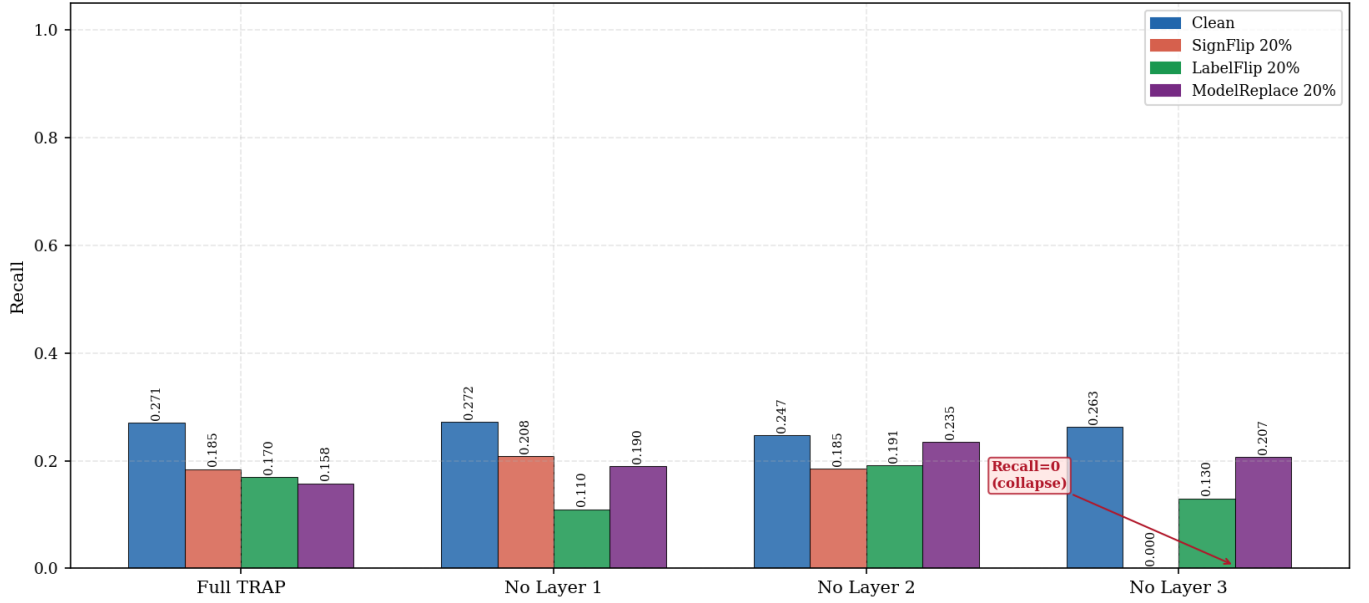


Fig. 10. Ablation study: final-round Recall. Removing Layer 3 under SignFlip 20% causes Recall=0 (all-benign collapse). Full TRAP recovers Recall=0.185, the highest among all configurations.

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TABLE IV
ABLATION STUDY: FINAL-ROUND AUC / F1 / RECALL

Config	Attack Condition			
	Clean	SF 20%	LF 20%	MR 20%
<i>AUC-ROC</i>				
Full TRAP	0.702	0.669	0.657	0.677
No Layer 1	0.705	0.686	0.578	0.665
No Layer 2	0.678	0.672	0.645	0.616
No Layer 3	0.700	0.808 [†]	0.571	0.650
<i>F1-Score</i>				
Full TRAP	0.409	0.304	0.284	0.270
No Layer 1	0.408	0.335	0.174	0.312
No Layer 2	0.385	0.305	0.246	0.367
No Layer 3	0.399	0.000 [†]	0.154	0.335
<i>Recall</i>				
Full TRAP	0.271	0.185	0.170	0.158
No Layer 1	0.272	0.208	0.110	0.190
No Layer 2	0.247	0.185	0.191	0.235
No Layer 3	0.263	0.000 [†]	0.130	0.207

[†] Collapse: F1=Recall=0; AUC reflects benign class only.